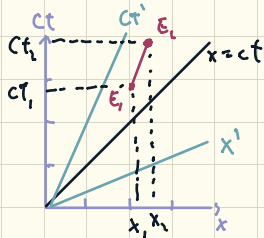


Demostración intervalo espacio-temporal



E_1 y E_2 tienen coordenadas (x_1, t_1) y (x_2, t_2) en el marco S , (x'_1, t'_1) y (x'_2, t'_2) en el marco S'

Definimos la cantidad $(\Delta s)^2$ con

$$(\Delta s)^2 = (c\Delta t)^2 - (\Delta x)^2 = (c(t_2 - t_1))^2 - (x_2 - x_1)^2 \quad (1)$$

Ahora lo hacemos en S'

$$(\Delta s')^2 = (c\Delta t')^2 - (\Delta x')^2 = (c(t'_2 - t'_1))^2 - (x'_2 - x'_1)^2 \quad (2)$$

Consideremos las transformadas de Lorentz

$$x'_i = \gamma(x_i - vt_i), \quad t'_i = \gamma(t_i - vx_i/c^2) \quad (3)$$

Reemplazamos en (1) en (2) Sabemos que $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$

$$\begin{aligned} (\Delta s')^2 &= (c\gamma[(t_2 - vx_2/c^2) - (t_1 - vx_1/c^2)])^2 - (\gamma[(x_2 - vt_2) - (x_1 - vt_1)])^2 \\ &= (c\gamma[t_2 - t_1 + \frac{v}{c^2}(x_1 - x_2)])^2 - (\gamma[x_2 - x_1 + v(t_1 - t_2)])^2 \\ &= (c\gamma[\Delta t - \frac{v}{c^2}\Delta x])^2 - (\gamma[\Delta x - v(\Delta t)])^2 \\ &= \gamma^2 [c^2\Delta t^2 - \cancel{2v\Delta t\Delta x} + \frac{v^2}{c^2}\Delta x^2 - \Delta x^2 + \cancel{2v\Delta x\Delta t} - v^2\Delta t^2] \\ &= \frac{1}{1 - v^2/c^2} \left[\Delta t^2 [c^2 - v^2] - \Delta x^2 [1 - \frac{v^2}{c^2}] \right] \\ &= \frac{1}{1 - v^2/c^2} \left(c^2\Delta t^2 [1 - \frac{v^2}{c^2}] - \Delta x^2 [1 - \frac{v^2}{c^2}] \right) \\ &= (c\Delta t)^2 - (\Delta x)^2 = (\Delta s)^2 \end{aligned}$$